

10-09-2023

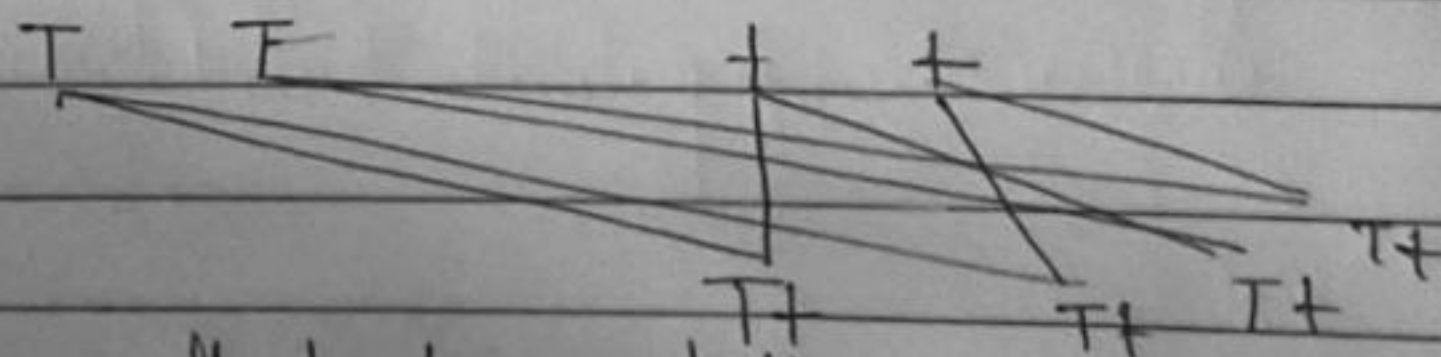
BIOLOGY

- Q.1 — (1) Tonoplast  
Q.2 — (4) Soluble starch  
Q.3 — (1) Haploid set of chromosomes  
Q.4 — (2)  $H_2O$   
Q.5 — (3)  $K^+$   
Q.6 — (4) Hydathodes  
Q.7 — (1) Telophase  
Q.8 — (3)  $As$  and  $gs$   
Q.9 — (3) Overall, water molecules moves from left to right  
Q.10 — (4) Hot, dry and windy  
Q.11 — (3) Stoma  
Q.12 — Asters (2)  
Q.13 — (2) Sex chromosomes  
Q.14 — (4) 25% TT, 50% Tt, 25% tt  
Q.15 — (4) Stop growing and die gradually of starvation

Q.16 Mendel's first law of inheritance are:-

→ Law of dominance: The first law of inheritance is law of dominance means which suppressed the recessive then get dominant over it and expressed own character.

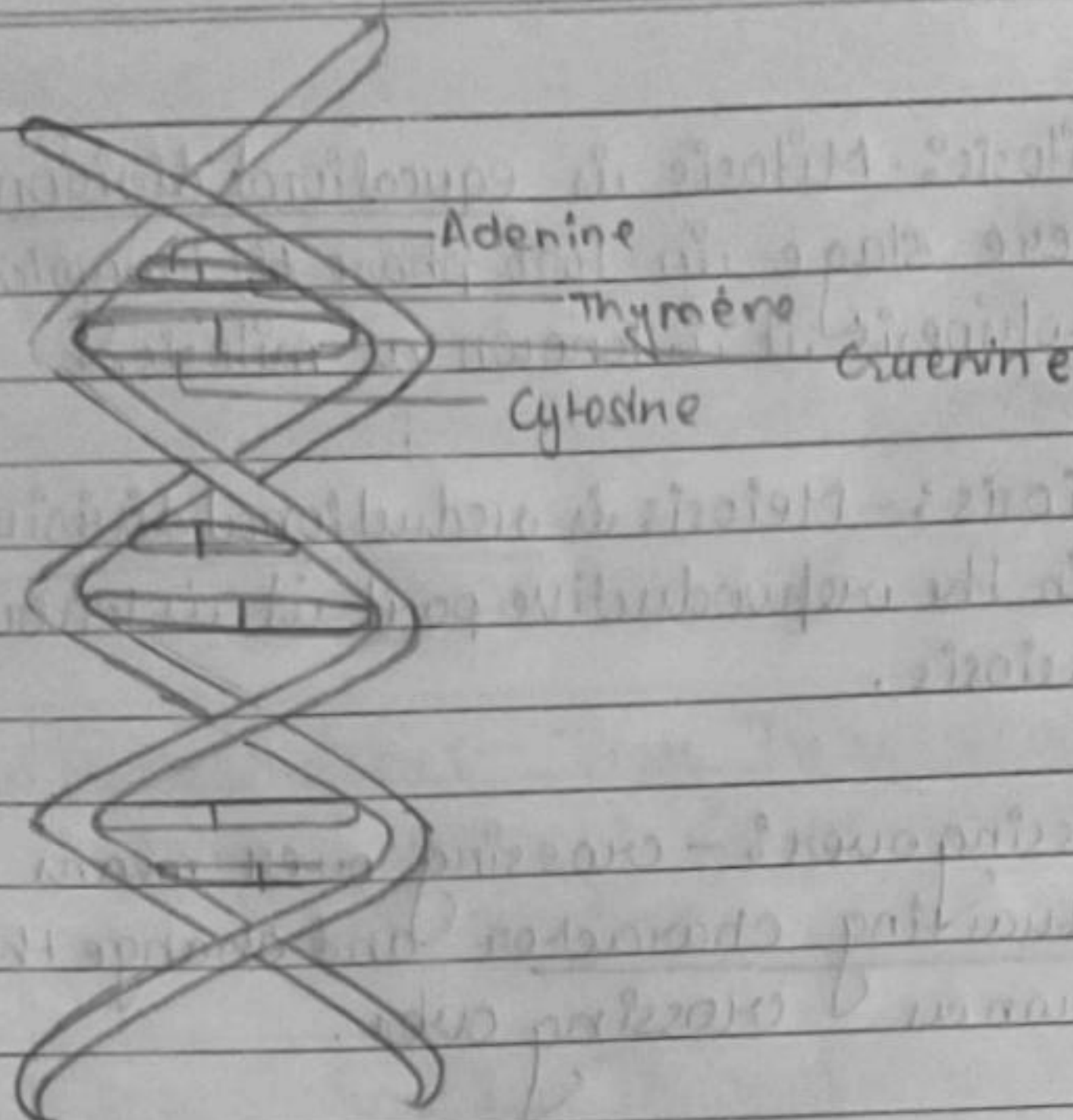
For example Tall Plant (TT) cross over short Plant (tt)



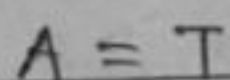
In  $F_1$  phase all plants are tall.



(ii)



The DNA (Double helical model) of four nitrogenous product Adenine, Thymine, Guanine and Cytosine. Whereas Adenine made double hydrogen bond with Thymine (T). Then, Guanine made triple hydrogen bond with Cytosine.



(iii) (a) The meaning of destarched plants is plant leaves without starch. It can destarched by the Iodo iodine solution.

(b) (i) Take a plants leave ~~with~~ <sup>the presence</sup> starch. without the presence of

(ii) Then, cover a half part of the leaf with <sup>starch</sup> the help of black cloth.

(iii) leave the plants at the light place.

(iv) Then, use iodine solution where doesn't not present starch it converted into Brown in colour.



(iv) The function of stomata are followed: -

- It regulates the exchanges of gases ( $\text{CO}_2$  and  $\text{O}_2$ )
- It maintain the concentration level of water.

(v) Three kind of transpiration are:-

- Stomatal transpiration: - Transpiration through minute opening of stomata in the leaf is known as stomatal transpiration.
- Cuticular transpiration: - Transpiration through cuticles present on the leaves is known as cuticular transpiration.
- Lenticular transpiration: - Transpiration through old bark tree present at the stem is known as lenticular transpiration.

\* minimum transpiration - lenticular

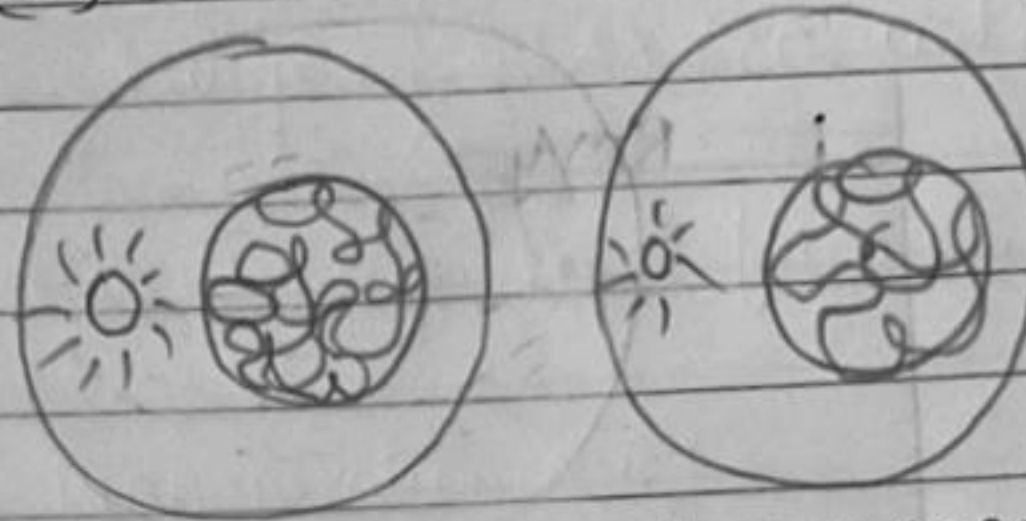
\* maximum transpiration - stomatal



Q.18 (i) Prophase

(ii) centromere, (2) chromatids, (3) spindle fibre

(iii)



nuclear membrane appears

(iv) (a) mitosis division

(b) meiosis division

| (v) | mitosis                                                                                                                                                                                 | meiosis                                                                                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <ul style="list-style-type: none"> <li>• 2 daughter cell produced</li> <li>• equational division</li> <li>• It present somatic and vegetative part.</li> <li>• It is diploid</li> </ul> | <ul style="list-style-type: none"> <li>• 4 daughter cell produced</li> <li>• reductional division</li> <li>• It present on reproductive part.</li> <li>• It is haploid</li> </ul> |



Q.21 (i) A - hypotonic solution  
B - hypertonic solution

(ii) A - deplasmolysis  
B - plasmolysis

(iii) hypotonic (cell A)  
hypertonic (cell B)

(iv) Osmotic pressure, root pressure

(v) The path of ascent of sap is :-

Cohesion :- Water molecules attached with water molecules

Adhesion :- water molecules attached with other substance.

root pressure :- The pressure exerted by the root for the movement of water molecules.